Air Data, Attitude and Heading Reference System (ADAHRS) (KSG 7200) – General Data

1. General

The aircraft is equipped with a dual channel KSG 7200 Air Data, Attitude and Heading Reference System (ADAHRS). The system provides primary attitude, heading and air data parameters from each channel to the Modular Avionics Unit (MAU).

2. Description

A. Air Data, Attitude and Heading Reference System (KSG 7200)

Each channel of the ADAHRS contains a solid-state micro-electro-mechanical system (MEMS) technology sensor block, which contains three rate sensors and three accelerometers in an orthogonal triad configuration. The triad in Channel B is skewed relative to Channel A. Each channel has a connection for an Outside Air Temperature (OAT) probe, a magnetometer, and two isolated absolute pressure sensors (one for pitot and one for static pressure). Channel A receives inputs from the left pitot tube, lower static ports, and one channel of the temperature probe and magnetometer. Channel B receives inputs from the right pitot tube, upper static ports, and the second channel of the temperature probe and magnetometer. Each channel of the ADAHRS has a Central Processing Unit (CPU). The ADAHRS is installed in the nose avionics rack. A Rosemount (Goodrich) dual element OAT sensor is mounted just forward and below the ram air scoop on the left side of the fuselage to supply the ADAHRS with OAT data. Two KMG 7010 magnetometers are mounted near the outboard end of the right wing (Pre Mod 6/2171), or in the vertical stabilizer (Post Mod 6/2171), one for each channel of the ADAHRS. During normal operation, the left pilot's PFD receives ADAHRS source data from the left pitot tube and lower static ports and ADAHRS Channel A. The right co-pilot's PFD receives ADAHRS source data from the right pitot tube, upper static ports, and ADAHRS Channel B. Each ADAHRS channel has a KCM 200 configuration module installed into the wiring harness near the KSG 7200 connector, which stores configuration and calibration data for the ADAHRS system.

Each ADAHRS channel CPU receives air data, temperature and heading information from that channel's sensor block and passes it to the other channel. Both CPUs then compare the data to verify sensor integrity. Verified ADAHRS information is sent to each channel of the MAU for display on the Apex System.

If the data from a sensor does not pass the verification check the data is discarded and will not be used. A fault signal will be sent to the MAU and a caution will be posted on the CAS. In this case the ADAHRS button on the PFD Controller of the failed side can be pressed to change the ADAHRS source channel to the opposite side. An amber text message will appear in the HSI window when both PFDs are using the same ADAHRS source channel.

28 Volt DC power is protected by a 3 Amp ADAHRS 1A (1K) (Left Bus) and a 3 Amp ADAHRS 1B (2K) (Right Bus) circuit breaker located on the avionics circuit breaker panel (refer to 24-00-00, Electrical Power – General Data).

The ADAHRS interfaces to the following aircraft systems:

1) Modular Avionics Unit (MAU) (refer to 31-43-00, Modular Avionics Unit (MAU) (Mod 6/2049) – General Data).

2) Traffic Alert and Collision Avoidance System (TCAS) (refer to 34-43-00, Traffic Alert and Collision Avoidance System I (TCAS I) – General Data and 34-43-01, Traffic Alert and Collision Avoidance System II (TCAS II) (Mod S.O.O. 6219) – General Data), if installed.

3) Enhanced Ground Proximity Warning System (EGPWS) (refer to 34-44-00, Enhance Ground Proximity Warning System (TAWS) (Mod 6/2049) – General Data), if installed.

4) Global Positioning System (GPS) (refer to 34-56-00, Global Positioning System (GPS) (Mod 6/2049) – General Data).

5) Weather Radar System (WXR) (refer to 34-41-00, Weather Radar System (Mod 6/2049) – General Data), if installed.

3. Adjustment/Test

Note The procedures in Para 3. Adjustment/Test are written to be stand-alone sub-sections. Each individual procedure includes aircraft and maintenance laptop computer set up and close out procedures. In cases where multiple procedures will be carried out in succession, it is acceptable to proceed to the subsequent procedure without the intermediate close out and set up procedure.

A. KSG 7200 ADAHRS Certification File Loading

1) General

The following procedures are for the loading of the KSG 7200 ADAHRS Certification File.

The KSG 7200 ADAHRS certification file change is required when the aircraft landing gear configuration is changed. Reloading the certification file will be required if the ADAHRS configuration modules are required to be replaced due to defect.

2) ADAHRS Certification File Loading

Note The KSG 7200 has two distinct channels, A and B. The following procedures must be done on both channels. The certification file loading procedure must be completed on both channels independently. Do the procedure on Channel A, then repeat for Channel B.

a) Ensure ADAHRS 1A (1K) and ADAHRS (1B) (2K) circuit breakers on the avionics circuit breaker panel are closed.

b) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

c) Wait for the aircraft electrical systems to initialize and the attitude flags to clear.

d) Turn on the maintenance laptop computer and boot up to the desktop.

e) Connect the maintenance laptop computer to the ADAHRS using an RS-232 maintenance cable from an available computer COM port to the ADAHRS 1A maintenance port on the aircraft maintenance panel located in the forward avionics bay. Start the KSG 7200 Maintenance program on the laptop.

f) Select appropriate COM port.

g) Click ‘Start Maintenance'.

h) Click ‘Aircraft Installation Setup'.

i) Click ‘Load KSG 7200 Certification File'. The field maintenance program should close and the ADAHRS loader program should open.

j) Insert the KSG 7200 Certification files CD-ROM into the laptop maintenance computer CD-ROM drive.

Note In some cases the certification file may be located on other media type such as local computer hard drive or removable media such as USB memory stick. In this case during the following steps select the correct file location as applicable.

k) Select the KSG 7200 located at the top of the program.

l) Select the Certification File button and click OK. The file select screen will appear.

m) Select the CD-ROM drive or alternate media containing the certification file(s).

CAUTION **IF THIS OPERATION IS CANCELLED, THE UNIT MUST BE RESTARTED BY REMOVING POWER FOR 5 SECONDS AND THEN REAPPLYING POWER.**

n) Select the configuration file (.cmf) and click open. An upload status bar will appear.

o) Click on OK when the certification file upload is complete.

p) Close the ADAHRS LOADER program by selecting EXIT.

q) Reset the ADAHRS computer by cycling the ADAHRS 1A (1K) and ADAHRS 1B (2K) circuit breakers on the avionics circuit breaker panel.

r) Wait for the KSG 7200 to re-initialize and the attitude flags to clear.

s) Start the KSG 7200 Maintenance program on the maintenance laptop computer.

t) Select appropriate COM port.

u) Click ‘Start Maintenance'.

v) Click ‘Part Numbers'.

w) Verify the “Certification File ID Number” displayed in the part number window matches the part number of the file loaded in previous steps.

x) Close the maintenance program on the maintenance laptop computer.

CAUTION **ALWAYS ENSURE THE ADAHRS MAINTENANCE PROGRAM ON THE MAINTENANCE LAPTOP COMPUTER IS CLOSED PRIOR TO DISCONNECTING THE RS-232 MAINTENANCE CABLES BETWEEN THE COMPUTER AND ADAHRS.**

y) Switch the RS-232 maintenance cable to the ADAHRS 1B Maintenance Port on the aircraft maintenance panel and repeat the above certification file loading procedure for Channel B.

z) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

aa) Disconnect and remove the maintenance laptop computer and RS-232 maintenance cable from the aircraft.

B. ADAHRS Alignment

1) General

The following procedures are for the attitude alignment of the ADAHRS system.

2) KSG 7200 ADAHRS Alignment

Note The KSG 7200 has two distinct channels, A and B. The following procedure must be done on both channels. The maintenance laptop computer can be connected to ADAHRS Channel A first and the procedure repeated for Channel B. Alternately a second RS-232 maintenance cable with a USB to RS-232 converter can be utilized for the second serial input to the maintenance laptop. Two separate instances of the KSG 7200 maintenance program must be running in Windows.

a) Ensure ADAHRS 1A (1K) and ADAHRS 1B (2K) circuit breakers on the avionics circuit breaker panel are closed.

b) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

c) Wait for the aircraft electrical systems to initialize and the attitude flags to clear.

d) Turn on the maintenance laptop computer and boot up to the desktop.

e) Connect the maintenance laptop computer to the ADAHRS using an ADAHRS 1A and/or ADAHRS 1B maintenance port on the aircraft maintenance panel located in the forward avionics bay.

f) Start the KSG 7200 Maintenance program on the maintenance laptop computer.

g) Select appropriate COM port.

h) Click ‘Start Maintenance'.

i) Click 'Aircraft Installation Setup'.

j) Click 'ADAHRS Alignment'.

Table 1 ADAHRS Pitch and Roll Offsets

Parameter		Setting
KSG 7200 ADAHRS Channel A and Channel B	Pitch Offset	+1.2
	Roll Offset	0
	Heading Offset	0
	Mounting Direction	ADAHRS Connector pointing towards Right Wing of Aircraft

k) Enter the Pitch Offset Value per Table 1 — ADAHRS Pitch and Roll Offsets.

l) Click ‘Set ADAHRS Pitch Offset'.

m) Enter the Roll Offset Value per Table 1 — ADAHRS Pitch and Roll Offsets.

n) Click ‘Set ADAHRS Roll Offset'.

o) Enter the Heading Offset Value per Table 1 — ADAHRS Pitch and Roll Offsets.

p) Click ‘Set ADAHRS Heading Offset'.

q) Click ‘Store ADAHRS Offsets'.

r) When ‘SUCCESS' is displayed, click ‘Close'.

s) Select the ADAHRS Mounting Direction number per Table 1 — ADAHRS Pitch and Roll Offsets.

t) Click ‘Set ADAHRS Mounting Direction'.

u) Click ‘Store ADAHRS Mounting Direction'.

v) When ‘SUCCESS' is displayed, click ‘Close' on the ADAHRS Mounting Installation screen.

w) Click ‘Exit Maintenance'.

x) Wait for the KSG 7200 to re-initialize and the attitude flags to clear.

y) Click ‘Start Maintenance'.

z) Click 'Aircraft Installation Setup'.

aa) Click 'ADAHRS Alignment'.

ab) Confirm Pitch Roll and Heading Offsets and Mounting Direction match the values entered above.

ac) Click 'Close' on the ADAHRS Mounting Installation screen.

ad) Click 'Exit Maintenance'. The ADAHRS will restart when the maintenance program is exited.

ae) Close the Maintenance Program on the maintenance laptop computer.

CAUTION **ALWAYS ENSURE THE ADAHRS MAINTENANCE PROGRAM ON THE MAINTENANCE LAPTOP COMPUTER IS CLOSED PRIOR TO DISCONNECTING THE RS-232 MAINTENANCE CABLES BETWEEN THE COMPUTER AND ADAHRS.**

af) If Channel B alignment procedure was not completed simultaneously with Channel A above, switch the RS-232 maintenance cable to the ADAHRS 1B Maintenance Port on the aircraft maintenance panel and repeat the above KSG 7200 alignment procedure for Channel B.

ag) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

ah) Disconnect and remove the maintenance laptop computer and RS-232 maintenance cables from the aircraft.

3) KMG 7010 Magnetometer Alignment (Pre Mod 6/2171)

a) Ensure that ADAHRS 1A (1K) and ADAHRS 1B (2K) circuit breakers on the avionics circuit breaker panel are closed.

b) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

c) Wait for the aircraft electrical systems to initialize and the attitude flags to clear.

d) Turn on maintenance laptop computer and boot up to the desktop.

e) Connect the maintenance laptop computer to the ADAHRS using an RS-232 maintenance cable from an available computer COM port to the ADAHRS 1A and/or ADAHRS 1B maintenance port on the aircraft maintenance panel located in the forward avionics bay.

f) Start the KSG 7200 Maintenance program on the laptop.

g) Select appropriate COM port.

h) Click 'Start Maintenance'.

i) Click 'Aircraft Installation Setup'.

j) Click 'Magnetometer Alignment'.

Table 2 Magnetometer Pitch and Roll Offsets

Parameter		Setting
KMG 7010 Magnetometers A and B	Pitch Offset	+2.2
	Roll Offset Normal Flight Operation	–3.3
	Roll Offset On-Ground Compass Calibration	–2.3
	Mounting Direction	Option “7” Bottom of magnetometer (mounting side of the unit) towards the top of aircraft. Front of magnetometer (connector side of the unit) towards tail of the aircraft.

k) Enter the Magnetometer Pitch Offset Value per Table 2 — Magnetometer Pitch and Roll Offsets.

l) Click ‘Set Magnetometer Pitch Offset'.

m) Enter the Magnetometer Roll Offset Value per Table 2 — Magnetometer Pitch and Roll Offsets.

Note If setting up for the compass calibration procedure, enter the Roll Offset for ‘On-Ground Compass Calibration'.

If setting up for flight, enter the Roll Offset for ‘Normal Flight Operation'.

n) Click ‘Set Magnetometer Roll Offset'.

o) Click ‘Store Magnetometer Offsets'.

p) When ‘SUCCESS' is displayed, click ‘Close'.

q) Select the Magnetometer Mounting Direction number per Table 2 — Magnetometer Pitch and Roll Offsets.

r) Click ‘Set Magnetometer Mounting Direction'.

s) Click ‘Store Magnetometer Mounting Direction'.

t) When ‘SUCCESS' is displayed, click ‘Close'.

u) Click ‘Close' on the Magnetometer Mounting Installation Screen.

v) Click ‘Close' on the Installation Setup Screen.

w) Click ‘Exit Maintenance'. The ADAHRS will restart when the maintenance program is exited.

x) Wait for the KSG 7200 to re-initialize and the attitude flags to clear.

y) Click ‘Start Maintenance'.

z) Click ‘Aircraft Installation Setup'.

aa) Click ‘Magnetometer Alignment'.

ab) Confirm pitch and roll offsets and Mounting Direction match the values entered above.

ac) Click ‘Close' on the Magnetometer Mounting Installation screen.

ad) Click ‘Exit Maintenance'. The ADAHRS will restart when the maintenance program is exited.

ae) Close the Maintenance Program on the maintenance laptop computer.

CAUTION **ALWAYS ENSURE THE ADAHRS MAINTENANCE PROGRAM ON THE MAINTENANCE LAPTOP COMPUTER IS CLOSED PRIOR TO DISCONNECTING THE RS-232 MAINTENANCE CABLES BETWEEN THE COMPUTER AND ADAHRS.**

af) If Channel B alignment procedure was not completed simultaneously with Channel A above, switch the RS-232 maintenance cable to the ADAHRS 1B Maintenance Port on the aircraft maintenance panel and repeat the above Magnetometer alignment procedure for Channel B.

ag) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

ah) Disconnect and remove the maintenance laptop computer and RS-232 maintenance cables from the aircraft.

4) KMG 7010 Magnetometer Alignment (Post Mod 6/2171)

a) Ensure that ADAHRS 1A (1K) and ADAHRS 1B (2K) circuit breakers on the avionics circuit breaker panel are closed.

b) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

c) Wait for the aircraft electrical systems to initialize and the attitude flags to clear.

d) Turn on maintenance laptop computer and boot up to the desktop.

e) Connect the maintenance laptop computer to the ADAHRS using an RS-232 maintenance cable from an available computer COM port to the ADAHRS 1A and/or ADAHRS 1B maintenance port on the aircraft maintenance panel located in the forward avionics bay.

f) Start the KSG 7200 Maintenance program on the laptop.

g) Select appropriate COM port.

h) Click ‘Start Maintenance'.

i) Click ‘Aircraft Installation Setup'.

j) Click ‘Magnetometer Alignment'.

Table 3 Magnetometer Pitch and Roll Offsets

Parameter		Setting
KMG 7010 Magnetometers A and B	Pitch Offset	0.0
	Roll Offset Normal Flight Operation	0.0
	Mounting Direction	Option “7” Bottom of magnetometer (mounting side of unit) towards the top of aircracft. Front of magnetometer (connector side of the unit) towards tail of the aircraft.

k) Enter the Magnetometer Pitch Offset Value per Table 3 — Magnetometer Pitch and Roll Offsets.

l) Click ‘Set Magnetometer Pitch Offset'.

m) Enter the Magnetometer Roll Offset Value per Table 3 — Magnetometer Pitch and Roll Offsets.

n) Click ‘Set Magnetometer Roll Offset'.

o) Click ‘Store Magnetometer Offsets'.

p) When ‘SUCCESS' is displayed, click ‘Close'.

q) Select the Magnetometer Mounting Direction number per Table 3 — Magnetometer Pitch and Roll Offsets.

r) Click ‘Set Magnetometer Mounting Direction'.

s) Click ‘Store Magnetometer Mounting Direction'.

t) When ‘SUCCESS' is displayed, click ‘Close'.

u) Click ‘Close' on the Magnetometer Mounting Installation Screen.

v) Click ‘Close' on the Installation Setup Screen.

w) Click ‘Exit Maintenance'. The ADAHRS will restart when the maintenance program is exited.

x) Wait for the KSG 7200 to re-initialize and the attitude flags to clear.

y) Click ‘Start Maintenance'.

z) Click 'Aircraft Installation Setup'.

aa) Click ‘Magnetometer Alignment'.

ab) Confirm pitch and roll offsets and Mounting Direction match the values entered above.

ac) Click ‘Close' on the Magnetometer Mounting Installation screen.

ad) Click ‘Exit Maintenance'. The ADAHRS will restart when the maintenance program is exited.

ae) Close the Maintenance Program on the maintenance laptop computer.

CAUTION **ALWAYS ENSURE THE ADAHRS MAINTENANCE PROGRAM ON THE MAINTENANCE LAPTOP COMPUTER IS CLOSED PRIOR TO DISCONNECTING THE RS-232 MAINTENANCE CABLES BETWEEN THE COMPUTER AND ADAHRS.**

af) If Channel B alignment procedure was not completed simultaneously with Channel A above, switch the RS-232 maintenance cable to the ADAHRS 1B Maintenance Port on the aircraft maintenance panel and repeat the above Magnetometer alignment procedure for Channel B.

ag) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

ah) Disconnect and remove the maintenance laptop computer and RS-232 maintenance cables from the aircraft.

C. ADAHRS System Function Test

1) General

The following procedures are for the function test of the ADAHRS System.

2) ADAHRS Function Test

Note The KSG 7200 has two distinct channels, A and B. The following procedure must be done on both channels. The maintenance laptop computer can be connected to ADAHRS Channel A first and the procedure repeated for Channel B. Alternately a second RS-232 maintenance cable with a USB to RS-232 converter can be utilized for the second serial input to the maintenance laptop. Two separate instances of the KSG 7200 maintenance program must be running in Windows.

a) Ensure ADAHRS 1A (1K) and ADAHRS 1B (2K) circuit breakers on the avionics circuit breaker panel are closed.

b) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

c) Wait for the aircraft electrical systems to initialize and the attitude flags to clear.

d) Verify correct display of ADAHRS parameters on pilot and co-pilot PFD's, including attitude, heading, airspeed, altitude, vertical speed, and OAT.

e) Turn on the maintenance laptop computer and boot up to the desktop.

f) Connect the maintenance laptop computer to the ADAHRS using an RS-232 maintenance cable from an available computer COM port to the ADAHRS 1A and/or ADAHRS 1B maintenance port on the aircraft maintenance panel located in the forward avionics bay.

g) Start the KSG 7200 Maintenance program on the maintenance laptop computer.

h) Select the appropriate COM port.

i) Click ‘Start Maintenance'.

j) Click ‘Built in Test Results'.

k) Verify all parameters indicate ‘Passed'.

l) Click ‘Close'.

m) Click ‘Exit Maintenance'. The ADAHRS will restart when the maintenance program is exited.

n) Close the Maintenance Program on the laptop maintenance computer.

CAUTION ALWAYS ENSURE THE ADAHRS MAINTENANCE PROGRAM ON THE MAINTENANCE LAPTOP COMPUTER IS CLOSED PRIOR TO DISCONNECTING THE RS-232 MAINTENANCE CABLES BETWEEN THE COMPUTER AND ADAHRS.

o) If Channel B ADAHRS System Function Test procedure was not completed simultaneously with Channel A above, switch the RS-232 maintenance cable to the ADAHRS 1B Maintenance Port on the aircraft maintenance panel and repeat the above ADAHRS Function Test procedure for Channel B.

p) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

q) Remove the maintenance laptop computer and RS-232 maintenance cables from the aircraft.

D. ADAHRS (KSG 7200) Air Data System Verification

1) General

The following procedures are for the Air Data System Verification of the ADAHRS System.

2) ADAHRS (KSG 7200) Air Data System Verification

a) Ensure that ADAHRS 1A (1K) and ADAHRS 1B (2K) circuit breakers on the avionics circuit breaker panel are closed.

b) Ensure that the 'PITOT HEAT' switch is off and the PITOT HEAT L (A6) and PITOT HEAT R (B6) circuit breakers on the main circuit breaker panel are pulled and fitted with safety clips.

c) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

d) Wait for the aircraft electrical systems to initialize and the attitude flags to clear.

e) Leak test the Pitot-Static System. (Refer to 34-11-00, Pitot-Static System (Mod 6/2033) – General Data, Adjustment/Test).

f) Ensure that the (ESIS) STATIC SOURCE valve is in the NORMAL position.

g) Connect the air data test set to the pitot heads and static ports on the aircraft. (Refer to 34-11-00, Pitot-Static System (Mod 6/2033) – General Data, Adjustment/Test).

h) Complete the Altitude Calibration Table. Refer to Table 4 — Altitude Calibration Table.

Note Additional maintenance and inspection requirements for the Altimeter System may be mandated by the regulations of the jurisdiction under which the aircraft is registered. In the event of disagreement between the PSM 1-64-2 and these additional requirements, the more restrictive requirements shall take precedence unless an alternative program of maintenance has been approved by the regulatory authority.

Table 4 Altitude Calibration Table

Test Set Altitude (feet)	Tolerance (± feet)	Record Error		REMARKS	INITIALS
		CH A	CH B		
-1,000	20				
0	20				
500	20				
1,000	20				
1,500	25				
2,000	25				
3,000	25				
4,000	25				
5,000	25				
6,000	25				
8,000	30				
10,000	30				

Table 4 Altitude Calibration Table (continued)

Test Set Altitude (feet)	Tolerance (± feet)	Record Error		REMARKS	INITIALS
		CH A	CH B		
12,000	35				
14,000	40				
16,000	40				
18,000	45				
20,000	50				
22,000	50				
25,000	50				

i) Complete the Airspeed Calibration Table. Refer to Table 5 — Airspeed Calibration Table

Note Additional maintenance and inspection requirements for the Airspeed System may be mandated by the regulations of the jurisdiction under which the aircraft is registered. In the event of disagreement between the PSM 1-64-2 and these additional requirements, the more restrictive requirements shall take precedence unless an alternative program of maintenance has been approved by the regulatory authority.

Table 5 Airspeed Calibration Table

Test Set Airspeed (knots)	Tolerance (± knots)	Record Error		REMARKS	INITIALS
		CH A	CH B		
30	7.0				
50	5.0				
80	3.5				
100	2.0				
120	2.0				
150	2.0				
200	2.0				

j) Disconnect the Pitot/Static Test Set from the pitot and static ports on the aircraft.

k) Ensure any tape has been removed and pitot and static ports are clear.

l) Remove the safety clips and reset the PITOT HEAT L (A6) and PITOT HEAT R (B6) circuit breakers on the main circuit breaker panel.

m) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

E. ADAHRS (KSG 7200) Compass Calibration & Verification

1) General

The following procedures are for the Compass Calibration and verification of the ADAHRS system.

2) ADAHRS System Compass Calibration (Pre Mod 6/2171)

Note The procedure below includes steps to adjust the magnetometer roll offset to compensate for wing flex arising from steady-state non-maneuvering flight. This wing flex can induce long-term compass heading errors, particularly in geographic areas with higher magnetic field dip angles (i.e., lower horizontal magnetic field strength). Heading errors induced by wing flex are reduced and considered negligible in geographic areas with small magnetic field dip angles as is the case for aircraft located in more equatorial regions. These considerations apply equally to northern and southern latitudes.

For best results the compass calibration procedure should be done on level ground without obstructions nearby. It is also recommended the calibration be done on a calm day. Excessive wind can rock the aircraft and result in compass errors, particularly in geographic locations with high magnetic field dip angles.

The compass calibration may be completed one channel at a time. However, more accurate results can be achieved by connecting both channels simultaneously. A second RS-232 maintenance cable with a USB to RS-232 converter can be utilized for the second serial input to the maintenance laptop. Two separate instances of the KSG 7200 maintenance program must be running in Windows.

a) Prior to compass calibration, ensure Magnetometer ‘On-Ground Compass Calibration Procedure' Roll Offset is entered in the KMG 7010 Magnetometer Alignment screen. Refer to Para 3.B.3) KMG 7010 Magnetometer Alignment (Pre Mod 6/2171).

b) Connect the maintenance laptop computer to the ADAHRS using an RS-232 maintenance cable from an available computer COM port to the ADAHRS 1A and/or ADAHRS 1B maintenance port on the aircraft maintenance panel located in the forward avionics bay.

Note The RS-232 maintenance cables for channel A and B can be routed from the maintenance panel to the flight compartment area, for convenience, for the calibration procedure.

c) Position aircraft on compass rose.

Note A ground support technician using a land sight compass with an accuracy of ± 2 degrees can be used if a compass rose is not available.

Note Less desirable and only if required due to circumstances compass calibration may also be done against a known calibrated on board compass source such as the offside ADAHRS Channel or ESIS instrument on condition the compass source has been verified by external means for accuracy prior to commencement of the following ADAHRS compass calibration procedures.

d) Ensure engines, avionics, and all normal electrical flight loads are operating for the duration of the compass calibration.

e) Turn on maintenance laptop computer and boot up to the desktop.

f) If not already completed, apply power to the ADAHRS. Wait for a minimum of 2 minutes and until the attitudes stabilize.

g) Start the KSG 7200 Maintenance program on the laptop.

h) Select appropriate COM port.

i) Click ‘Start Maintenance'.

j) Click ‘Aircraft Installation Setup'.

k) Click ‘Compass Calibration'.

l) Click ‘Perform Compass Calibration using 12 points on the Compass Rose'.

m) Align the aircraft to the heading requested in the compass rose calibration window.

n) **IMPORTANT** Wait a minimum of 1 minute to let the ADAHRS system stabilize at each new heading.

o) Click ‘Take Magnetometer Reading' when the aircraft is aligned to the requested heading.

p) Repeat the compass calibration at all 12 points observing the minimum 1 minute to stabilize at all headings.

q) Upon completion of all 12 heading calibrations the ‘Store Compass Rose Calibration Data' button becomes active.

r) Click ‘Store Compass Rose Calibration Data'.

s) When ‘SUCCESS' is displayed, click ‘Close'.

t) Click ‘Close'.

u) Click ‘Exit Maintenance'. The ADAHRS will restart when the maintenance program is exited.

v) Close the Maintenance Program on the maintenance laptop computer.

w) Keep aircraft stationary for a minimum of 2 minutes before proceeding.

CAUTION **ALWAYS ENSURE THE ADAHRS MAINTENANCE PROGRAM ON THE MAINTENANCE LAPTOP COMPUTER IS CLOSED PRIOR TO DISCONNECTING THE RS-232 MAINTENANCE CABLES BETWEEN THE COMPUTER AND ADAHRS.**

x) If Channel B alignment procedure was not completed simultaneously with Channel A above, switch the RS-232 maintenance cable to the ADAHRS 1B Maintenance Port on the aircraft maintenance panel and repeat the above compass calibration procedure for Channel B.

y) Do the compass accuracy verification, outlined in Para 3.E.4).

3) ADAHRS System Compass Calibration (Post Mod 6/2171)

Note For best results the compass calibration procedure should be done on level ground without obstructions nearby. It is also recommended the calibration be done on a calm day. Excessive wind can rock the aircraft and result in compass errors, particularly in geographic locations with high magnetic field dip angles.

The compass calibration may be completed one channel at a time. However, more accurate results can be achieved by connecting both channels simultaneously. A second RS-232 maintenance cable with a USB to RS-232 converter can be utilized for the second serial input to the maintenance laptop. Two separate instances of the KSG 7200 maintenance program must be running in Windows.

a) Connect the maintenance laptop computer to the ADAHRS using an RS-232 maintenance cable from an available computer COM port to the ADAHRS 1A and/or ADAHRS 1B maintenance port on the aircraft maintenance panel located in the forward avionics bay.

Note The RS-232 maintenance cables for channel A and B can be routed from the maintenance panel to the flight compartment area, for convenience, for the calibration procedure.

b) Position aircraft on compass rose.

Note A ground support technician using a land sight compass with an accuracy of ± 2 degrees can be used if a compass rose is not available.

Note Less desirable and only if required due to circumstances compass calibration may also be done against a known calibrated on board compass source such as the offside ADAHRS Channel or ESIS instrument on condition the compass source has been verified by external means for accuracy prior to commencement of the following ADAHRS compass calibration procedures.

c) Ensure engines, avionics, and all normal electrical flight loads are operating for the duration of the compass calibration.

d) Turn on maintenance laptop computer and boot up to the desktop.

e) If not already completed, apply power to the ADAHRS. Wait for a minimum of 2 minutes and until the attitudes stabilize.

f) Start the KSG 7200 Maintenance program on the laptop.

g) Select appropriate COM port.

h) Click ‘Start Maintenance'.

i) Click ‘Aircraft Installation Setup'.

j) Click ‘Compass Calibration'.

k) Click ‘Perform Compass Calibration using 12 points on the Compass Rose'.

l) Align the aircraft to the heading requested in the compass rose calibration window.

m) **IMPORTANT** Wait a minimum of 1 minute to let the ADAHRS system stabilize at each new heading.

n) Click ‘Take Magnetometer Reading' when the aircraft is aligned to the requested heading.

o) Repeat the compass calibration at all 12 points observing the minimum 1 minute to stabilize at all headings.

p) Upon completion of all 12 heading calibrations the ‘Store Compass Rose Calibration Data' button becomes active.

q) Click ‘Store Compass Rose Calibration Data'.

r) When ‘SUCCESS' is displayed, click ‘Close'.

s) Click ‘Close'.

t) Click ‘Exit Maintenance'. The ADAHRS will restart when the maintenance program is exited.

u) Close the Maintenance Program on the maintenance laptop computer.

v) Keep aircraft stationary for a minimum of 2 minutes before proceeding.

CAUTION **ALWAYS ENSURE THE ADAHRS MAINTENANCE PROGRAM ON THE MAINTENANCE LAPTOP COMPUTER IS CLOSED PRIOR TO DISCONNECTING THE RS-232 MAINTENANCE CABLES BETWEEN THE COMPUTER AND ADAHRS.**

w) If Channel B alignment procedure was not completed simultaneously with Channel A above, switch the RS-232 maintenance cable to the ADAHRS 1B Maintenance Port on the aircraft maintenance panel and repeat the above compass calibration procedure for Channel B.

x) Do the compass accuracy verification, outlined in Para 3.E.4).

4) Compass Accuracy Verification

a) Prior to compass accuracy verification (Pre Mod 6/2171), ensure Magnetometer On-Ground Compass Calibration Procedure Roll Offset is entered in the KMG 7010 Magnetometer Alignment maintenance screen. Refer to Para 3.B.3) KMG 7010 Magnetometer Alignment (Pre Mod 6/2171).

b) Ensure engines, avionics and all normal electrical flight loads are operating for the duration of the compass verification.

c) Align the aircraft to each of the four cardinal headings on the compass rose and verify compass accuracy ± 2.0 degrees to actual aircraft heading.

Note Additional maintenance and inspection requirements for the Compass System may be mandated by the regulations of the jurisdiction under which the aircraft is registered. In the event of disagreement between the PSM 1-64-2 and these additional requirements, the more restrictive requirements shall take precedence unless an alternative program of maintenance has been approved by the regulatory authority.

d) If any heading falls outside the listed specifications, do the compass calibration procedure listed in Para 3.E.2) (Pre Mod 6/2171) or Para 3.E.3) (Post Mod 6/2171).

5) Wing Flex Compensation (Pre Mod 6/2171)

a) Following compass accuracy verification procedure, compensate for in-flight wing flex by entering the Magnetometer Normal Flight Operation Roll Offset into the KMG 7010 Magnetometer Alignment maintenance screen. Refer to Para 3.B.3) KMG 7010 Magnetometer Alignment (Pre Mod 6/2171).

Note Compensating for in-flight wing flex by adjusting the Magnetometer Roll Offset following compass calibration will have varying effect on calibrated compass headings based on geographic location and magnetic field dip angle. Most effect will be noticed in Northern or Southern latitudes above/below 45 degrees latitude with high magnetic field dip angles indicated by up to 2 or 3 degree change in ground compass calibration readings. but this is required to ensure in-flight compass accuracy. Wing flex will have minimal effect in equatorial geographic areas with low magnetic field dip angles, but the wing flex compensation procedure should still be completed to ensure best in-flight compass performance.

F. Configuration Module Programming

1) General

The following procedures are for programming a new configuration module.

2) Configuration Module Programming

Note The following procedure can only be accomplished if the ADAHRS was previously installed and operating in the aircraft.

a) Ensure ADAHRS 1A (1K) and ADAHRS 1B (2K) circuit breakers on the avionics circuit breaker panel are closed.

b) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

c) Wait for the aircraft electrical systems to initialize and the attitude flags to clear.

d) Turn on maintenance laptop computer and boot up to the desktop.

e) Connect the maintenance laptop computer to the ADAHRS using an RS-232 maintenance cable from an available computer COM port to the ADAHRS 1A or ADAHRS 1B (as required) maintenance port on the aircraft.

f) Start the KSG 7200 Maintenance program on the maintenance laptop computer.

g) Select appropriate COM port.

h) Click ‘Start Maintenance'.

i) Click ‘NVM/Configuration Module Data Transfer'.

j) Click ‘Copy NVM Data to Configuration Module'.

k) Wait 60 seconds to assure transfer is complete.

l) Close the maintenance program on the maintenance laptop computer.

m) Wait for the ADAHRS to initialize and the attitude flags to clear.

n) Start the KSG 7200 Maintenance program on the maintenance laptop computer.

o) Select appropriate COM port.

p) Click ‘Start Maintenance'.

q) Click ‘Part Numbers'.

r) Verify valid data is present.

s) Close the maintenance program on the maintenance laptop computer.

CAUTION **ALWAYS ENSURE THE ADAHRS MAINTENANCE PROGRAM ON THE MAINTENANCE LAPTOP COMPUTER IS CLOSED PRIOR TO DISCONNECTING THE RS-232 MAINTENANCE CABLES BETWEEN THE COMPUTER AND ADAHRS.**

t) If the other configuration module also requires programming, switch the RS-232 maintenance cable to the ADAHRS 1A or ADAHRS 1B Maintenance Port (as applicable) on the aircraft maintenance panel and repeat the above configuration module programming procedure.

u) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

v) Remove the maintenance laptop computer and RS-232 maintenance cables from the aircraft.

4. Removal/Installation

WARNING **INSTALLATION OF THIS COMPONENT CONTAINS PROTECTION FEATURES THAT ARE CRITICAL TO PREVENTING DEGRADATION FROM THE EFFECTS OF HIGH INTENSITY RADIATED FIELDS (HIRF) AND INDIRECT EFFECTS OF LIGHTNING (IEL). ENSURE INTEGRITY OF THE HIRF/IEL PROTECTION FEATURES ARE MAINTAINED IN ACCORDANCE WITH THE FOLLOWING:**

CHASSIS GROUNDING – ALL RACK MOUNT / REMOTE MOUNT UNITS SHALL BE ELECTRICALLY BONDED TO THE AIRFRAME. ENSURE THAT THE MATING SURFACES BETWEEN THE LRU MOUNTING TRAY (OR LRU MOUNTING FEET IF A TRAY IS NOT USED) PROVIDE A LOW IMPEDANCE ELECTRICAL PATH. ALL MATING SURFACES MUST BE FREE OF ALL PAINT AND OTHER NONCONDUCTIVE ELEMENTS. REFER TO BONDING REQUIREMENTS IN 20-75-00, ELECTRICAL BONDING AND GROUNDING OF AIRCRAFT STRUCTURES – GENERAL DATA.

SHIELD GROUNDS – VERIFY SHIELD GROUNDING AT CONNECTOR BACKSHELLS AS PER REQUIREMENTS IN PSM 1-64-2W.

BACKSHELLS OR STRAIN RELIEFS – VERIFY PROPER ELECTRICAL BONDING BETWEEN THE BACKSHELL/STRAIN RELIEF AND THE MOUNTING TRAY OR LRU CASE AS PER REQUIREMENTS IN PSM 1-64-2W.

A. Removal and Installation of the ADAHRS (KSG 7200)

1) General

The following procedures are for the removal and installation of the ADAHRS (KSG 7200).

2) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Pull the ADAHRS 1A (1K) (Left Bus) and ADAHRS 1B (2K) (Right Bus) circuit breakers on the avionics circuit breaker panel and fit safety clips.

c) Gain access to the forward avionics shelves through the forward baggage compartment.

d) Remove middle cover attached to bulkhead station 37.21.

e) Disconnect the Pitot and Static lines from the front of the unit.

f) Disconnect the electrical connectors from the front of the unit.

g) Loosen retaining screw clamp from the front and slide the unit out of the mounting tray.

h) Remove unit from aircraft.

Note Stow connectors of the removed components. Install the pitot and static line covers.

i) Visually inspect mounting tray. If damaged replace as per Para 4.B.

3) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure that the ADAHRS 1A (1K) (Left Bus) and ADAHRS 1B (2K) (Right Bus) circuit breakers on the avionics circuit breaker panel are pulled and fitted with safety clips.

c) Slide the unit into the mounting tray. Tighten the two retaining screw clamps at the front lip of the ADAHRS unit.

d) Reconnect the electrical connector to the unit. Ensure connections are secure. Use lockwire .025 to secure the two retaining screws on channel A and B ADAHRS connectors.

e) Ensure Pitot and Static lines are free of debris. Reconnect the Pitot and Static lines. Ensure connections are secure.

f) Do a leak check on the Pitot and Static system as per 34-11-00, Pitot-Static System (Mod 6/2033) – General Data.

g) Install middle cover attached to bulkhead station 37.21.

WARNING **ACCESS COVER IS REQUIRED TO MAINTAIN PROTECTION FEATURES THAT ARE ESSENTIAL TO MINIMIZING ADVERSE EFFECTS FROM HIGH INTENSITY RADIATED FIELDS (HIRF).**

h) Remove safety clips and close ADAHRS 1A (1K) (Left Bus) and ADAHRS 1B (2K) (Right Bus) circuit breakers on the avionics circuit breaker panel.

i) Test the ADAHRS as per Para 3.C and Para 3.D.

B. Removal and Installation of the ADAHRS Mounting Tray

1) Removal

a) Remove ADAHRS as per Para 4.A.2).

b) Remove the four screws that secure mounting tray to shelf.

c) Remove mounting tray from aircraft.

2) Installation

a) Locate mounting tray on avionic shelf, align mounting holes. Ensure the tray in oriented properly on the shelf, hold down bolts are inboard.

b) Install the four screws and ensure tray is securely held on shelf.

C. Removal and Installation of the Magnetometers (KMG 7010)

1) General

The following procedures are for the removal and installation of the magnetometers (KMG 7010).

CAUTION **ONLY NON-MAGNETIC TOOLS ARE PERMITTED WITHIN 12 INCHES OF THE MAGNETOMETER.**

2) Removal (Pre Mod 6/2171)

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Pull the ADAHRS 1A (1K) (Left Bus) and ADAHRS 1B (2K) (Right Bus) circuit breakers on the avionics circuit breaker panel and fit safety clips.

c) Locate the panel on the starboard (RH) wing between station 272.5 and station 285.0.

d) Remove the (8) screws securing the panel to the wing with non-magnetic tools.

Note ADAHRS channel A magnetometer is mounted inboard, channel B outboard.

e) Remove the four mounting screws with non-magnetic tools.

f) Disconnect the electrical connector from the aft side of the unit.

g) Remove the unit from the aircraft.

Note Stow connectors of the removed components.

Note Store the magnetometer in a secure area, away from magnetic fields. Typically, a distance of 12 inches is considered safe for most magnetic fields but actual safe distance varies with the strength of the magnetic source. Exposing the magnetometer to any magnetic fields during transport, storage, re-packaging, or installation may disrupt its calibration. To minimize exposure of the magnetometer to strong magnetic fields, keep it away from the following devices or objects:

1 High Current Carrying Cables

2 Magnetic Tools or Fasteners

3 Permanent Magnets

4 Cathode Ray Tubes / Computer Monitors

5 Magnetic Surfaces

6 Motors & Servos

3) Installation (Pre Mod 6/2171)

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure that the ADAHRS 1A (1K) (Left Bus) and ADAHRS 1B (2K) (Right Bus) circuit breakers on the avionics circuit breaker panel are pulled and fitted with safety clips.

c) Reconnect the electrical connector to the unit. Ensure connections are secure. Use lockwire .025 to secure the two retaining screws on the connector.

d) Set unit in position and install the four mounting screws with non-magnetic tools.

e) Verify the electrical conductivity between the magnetometer and airframe is < 2.5 milliohms.

f) Install the panel screws (8) and secure to the wing with non-magnetic tools.

g) Remove safety clips and close ADAHRS 1A (1K) (Left Bus) and ADAHRS 1B (2K) (Right Bus) circuit breakers on the avionics circuit breaker panel.

h) Do a function test of the ADAHRS as per Para 3.C.

i) Do Compass Calibration & Verification procedure per Para 3.E.2) and Para 3.E.4).

j) Do Wing Flex Compensation per Para 3.E.5).

4) Removal (Post Mod 6/2171)

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Pull the ADAHRS 1A (1K) (Left Bus) and ADAHRS 1B (2K) (Right Bus) circuit breakers on the avionics circuit breaker panel and fit safety clips.

c) Remove fairings enclosing vertical stabilizer and horizontal stabilizer to rear fuselage. Refer to 12-00-00, Servicing – General Data, Figure 1, Access and Inspection Provisions.

Note ADAHRS channel A magnetometer is mounted on top, directly to mounting tray, channel B magnetometer is mounted below, using spacers.

d) Remove the four mounting screws, and four spacers with non-magnetic tools.

e) Disconnect the electrical connector from the aft side of the each unit.

f) Remove the units from the aircraft.

Note Stow connectors of the removed components.

Note Store the magnetometer in a secure area, away from magnetic fields. Typically, a distance of 12 inches is considered safe for most magnetic fields but actual safe distance varies with the strength of the magnetic source. Exposing the magnetometer to any magnetic fields during transport, storage, re-packaging, or installation may disrupt its calibration. To minimize exposure of the magnetometer to strong magnetic fields, keep it away from the following devices or objects:

1 High Current Carrying Cables

2 Magnetic Tools or Fasteners

3 Permanent Magnets

4 Cathode Ray Tubes / Computer Monitors

5 Magnetic Surfaces

6 Motors & Servos

5) Installation (Post Mod 6/2171)

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure that the ADAHRS 1A (1K) (Left Bus) and ADAHRS 1B (2K) (Right Bus) circuit breakers on the avionics circuit breaker panel are pulled and fitted with safety clips.

c) Reconnect the electrical connector to the unit. Ensure connections are secure. Use lockwire .025 to secure the two retaining screws on the connector.

d) Hold units in position on mounting tray.

Note ADAHRS channel A magnetometer is mounted on top, directly to mounting tray, channel B magnetometer is mounted below, using spacers.

e) Install four spacers, and four mounting screws with non-magnetic tools.

f) Verify the electrical conductivity between the magnetometer and airframe is < 2.5 milliohms.

g) Install fairings enclosing vertical stabilizer and horizontal stabilizer to rear fuselage.

h) Remove safety clips and close ADAHRS 1A (1K) (Left Bus) and ADAHRS 1B (2K) (Right Bus) circuit breakers on the avionics circuit breaker panel.

i) Do a function test of the ADAHRS as per Para 3.C.

j) Do Compass Calibration & Verification procedure per Para 3.E.3) and Para 3.E.4).

D. Removal and Installation of the Outside Air Temperature (OAT) Probe

1) General

The following procedures are for the removal and installation of the Outside Air Temperature (OAT) probe.

2) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Pull the ADAHRS 1A (1K) (Left Bus) and ADAHRS 1B (2K) (Right Bus) circuit breakers on the avionics circuit breaker panel and fit safety clips.

c) Carefully cut the sealant from around the base of the probe.

d) Remove the screws (6) from the base of the probe.

e) Carefully lift the probe clear of the fuselage until there is access to the electrical connector.

f) Disconnect the electrical connector.

Note Stow connectors of the removed components.

3) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure that the ADAHRS 1A (1K) (Left Bus) and ADAHRS 1B (2K) (Right Bus) circuit breakers on the avionics circuit breaker panel are pulled and fitted with safety clips.

c) Reconnect the electrical connector to the unit. Ensure connection is secure.

d) Install the screws (6) securing the probe to the fuselage.

e) Solvent clean the exposed area of the airframe around the probe (refer to 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data).

f) Touch up the exposed area of airframe around the probe if required with F19 primer by brush within 24 hours of the probe installation. Ensure that the primer re-finishing provides complete coverage of the affected area.

g) Use PR 1422 sealant to both fillet seal around the probe base and dome seal the securing screws. The aircraft will be ready to fly approximately 24 hours after sealing.

h) Remove safety clips and close ADAHRS 1A (1K) (Left Bus) and ADAHRS 1B (2K) (Right Bus) circuit breakers on the avionics circuit breaker panel.

i) Do the ADAHRS System Function Test procedure as per Para 3.C.

E. Removal and Installation of the Outside Air Temperature (OAT) Probe (S.O.O. 6304)

1) General

The following procedures are for the removal and installation of the Outside Air Temperature (OAT) probe.

2) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Pull the ADAHRS 1A (1K) (Left Bus) and ADAHRS 1B (2K) (Right Bus) circuit breakers on the avionics circuit breaker panel and fit safety clips.

c) Carefully cut the sealant from around the base of the probe.

d) Remove the screws (6) from the base of the probe.

e) Carefully cut fay seal between base of probe and aircraft skin and lift the probe clear of the fuselage until there is access to the electrical connector.

f) Disconnect the electrical connector.

Note Stow connectors of the removed components.

3) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure that the ADAHRS 1A (1K) (Left Bus) and ADAHRS 1B (2K) (Right Bus) circuit breakers on the avionics circuit breaker panel are pulled and fitted with safety clips.

c) Ensure leftover sealant is carefully removed from the base of the probe and the aircraft skin.

d) Reconnect the electrical connector to the unit. Ensure connection is secure.

e) Apply PR-1422 sealant to faying surface of the probe base and aircraft skin.

f) Install the screws (6) securing the probe to the fuselage.

g) Do a bonding check between the probe and aircraft structure (refer to 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data).

h) Touch up the exposed area of airframe used for a bonding check with alodine and F19 primer, by brush, within 24 hours of the probe installation. Ensure that the primer re-finishing provides complete coverage of the affected area.

i) Use PR-1422 sealant to both fillet seal around the probe base and dome seal the securing screws. The aircraft will be ready to fly approximately 24 hours after sealing.

j) Remove safety clips and close ADAHRS 1A (1K) (Left Bus) and ADAHRS 1B (2K) (Right Bus) circuit breakers on the avionics circuit breaker panel.

k) Do the ADAHRS System Function Test procedure as per Para 3.C.

F. Removal and Installation of the KCM 200 Configuration Modules

1) General

The following procedures are for the removal and installation of the KCM 200 Configuration Modules.

2) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Pull the ADAHRS 1A (1K) (Left Bus) and ADAHRS 1B (2K) (Right Bus) circuit breaker on the avionics circuit breaker panel and fit safety clips.

c) Gain access to the forward avionics shelves through the forward baggage compartment.

d) Remove middle cover attached to bulkhead station 37.21.

e) Remove electrical connector from the applicable channel of the ADAHRS.

f) Remove backshell from electrical connector and extract the configuration module pins from connector. Refer to ADAHRS wiring diagram for pinout information.

g) Cut cable ties as required to free the configuration module from aircraft harness.

Note Stow connectors of the removed components.

3) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure the ADAHRS 1A (1K) (Left Bus) and ADAHRS 1B (2K) (Right Bus) circuit breakers on the avionics circuit breaker panel are pulled and fitted with safety clips.

c) Prepare the configuration module harness as required to attach to the ADAHRS connector harness. Insert the configuration module pins into the connector. Refer to ADAHRS wiring diagram.

d) Install backshell onto connector and reconnect electrical connector to ADAHRS

e) Secure the configuration module into aircraft harness using cable ties, as required.

f) If a new KCM 200 Configuration Module is installed, programming is required:

1 If the original ADAHRS unit is installed, the configuration module can be programmed by the ADAHRS, refer to Para 3.F.

2 If the original ADAHRS is not installed, the configuration module will need to be programmed per Para 3.A, Para 3.B, and Para 3.E. Once programming is complete, do a function test per Para 3.C.

5. Troubleshooting

A. Air Data and Attitude Heading Reference System (KSG 7200)

1) If the Air Data and Attitude Heading Reference System (KSG 7200) is deemed inoperative check the following:

a) Ensure ADAHRS 1A (1K) (Left Bus) and the ADAHRS 1B (2K) (Right Bus) circuit breakers on the avionics circuit breaker panel are closed.

b) Check that the units are securely installed and the locking screw is securely tightened.

c) Visually inspect ADAHRS system connectors for damaged wires or contacts.

d) If troubleshooting at the wire level is required refer to the applicable wiring diagrams in Wiring Manual PSM 1-64-2W.

e) If repair or service is required, send to appropriate rated avionics repair facility.

6. Servicing

It is recommended that a compass accuracy verification be done (refer to Para 3.E.4)) whenever any of the following occur:

- Accuracy of the compass is suspected
- A modification or major replacement involving ferrous metals
- After lightning strike
- Whenever a modification or new installation is made to the aircraft electrical system

- When an aircraft operation is changed to a different geographical location with a major change in magnetic deviation and/or magnetic dip angle
- When magnetometers are replaced

Note Additional maintenance and inspection requirements for the installation of the ADAHRS may be mandated by the regulations of the jurisdiction under which the aircraft is registered. In the event of disagreement between the PSM 1-64-2 and these additional requirements, the more restrictive requirements shall take precedence unless an alternative program of maintenance has been approved by the regulatory authority.

A. Magnetometers (KMG 7010)

1) Magnetometer operation is tested as per Para 3.E.4) – Compass Accuracy Verification.

B. Outside Air Temperature (OAT) Probe

1) Outside Air Temperature (OAT) probe operation is tested as per Para 3.C – ADAHRS System Function Test.